



SC1006

Computer Organisation and Architecture

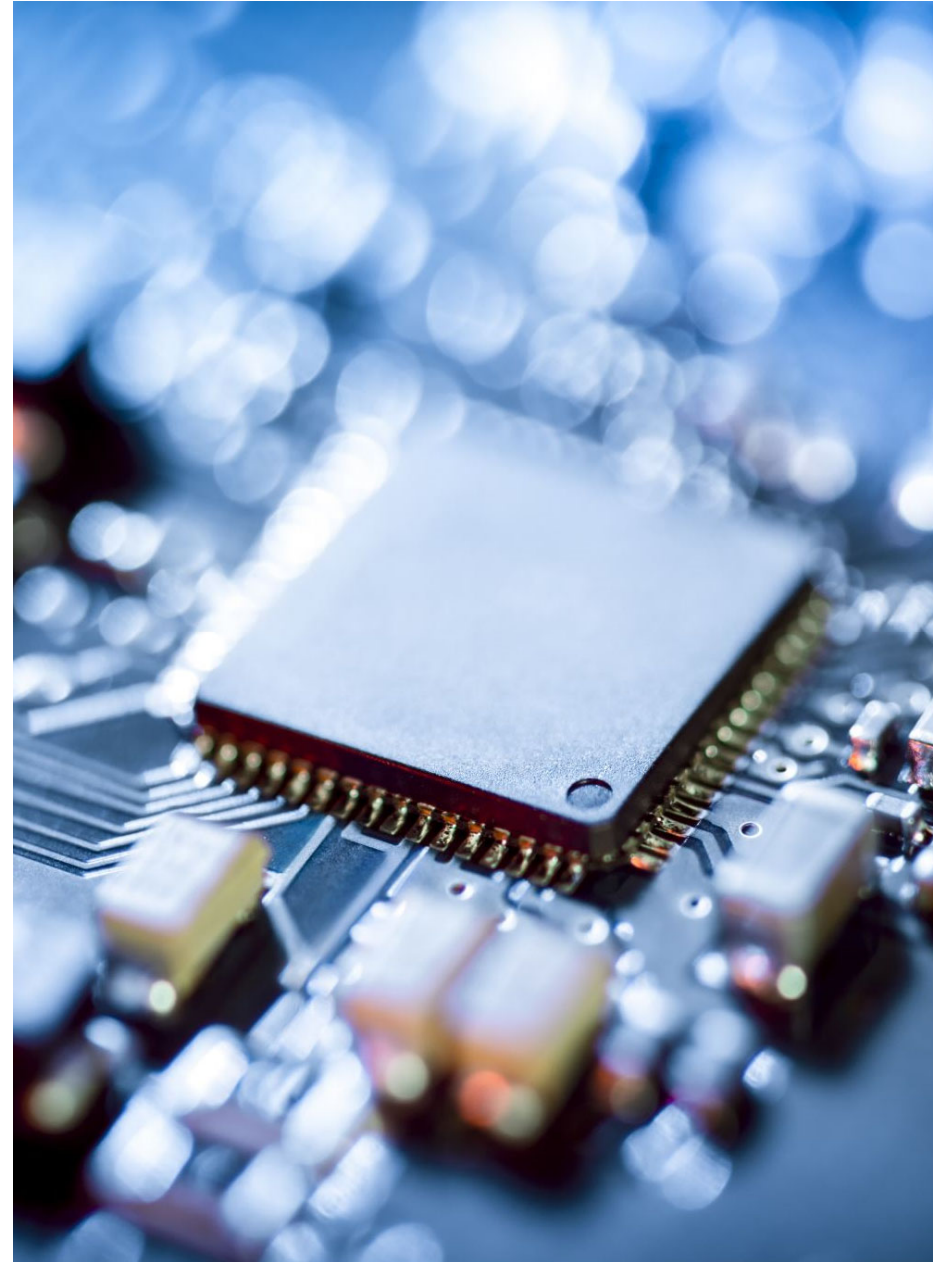
Volatile Memory

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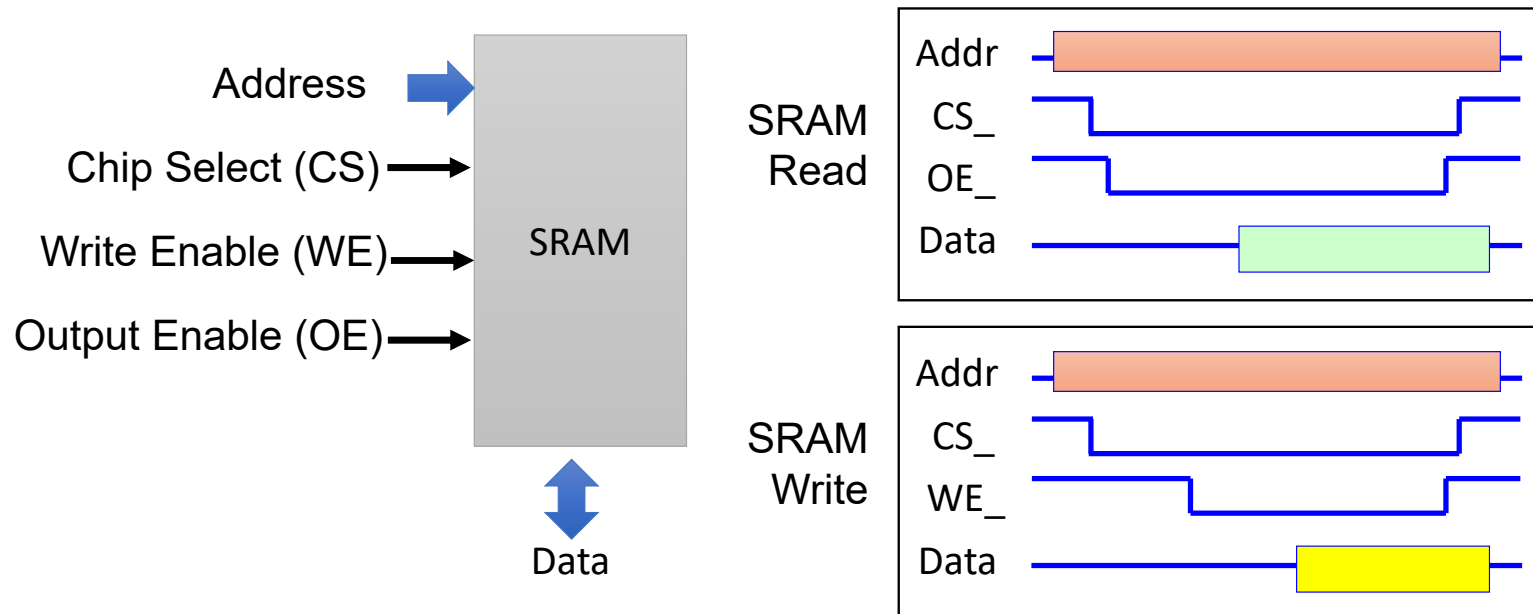


Random Access Memory (RAM)

- Static RAM (SRAM)
 - Static Random Access Memory
 - Data stored as long as supply is applied.
 - Large (4 to 6 transistors per cell).
 - Fast (SRAM is optimized for speed/access-latency)
 - Low power consumption (active and standby)
- Dynamic Random Access Memory (DRAM)
 - Periodic refresh required.
 - Small (1 to 3 transistors per cell).
 - Slower (DRAM is optimized for density/higher-capacity; refresh takes time)
 - Higher power consumption due to need for periodic refresh operation to maintain data integrity of memory cells.



Static RAM (SRAM) Access



Addr	Specifies Address of memory location.	
Data	Data to be read/written.	Typically 8,16 or 32 bits.
CS	Control signals	Chip Select (Enables or disables the chip).
WE		Write Enable (Allows data to to be written)
OE		Output Enable (Allows SRAM to output data)

SRAM Chip DeCap

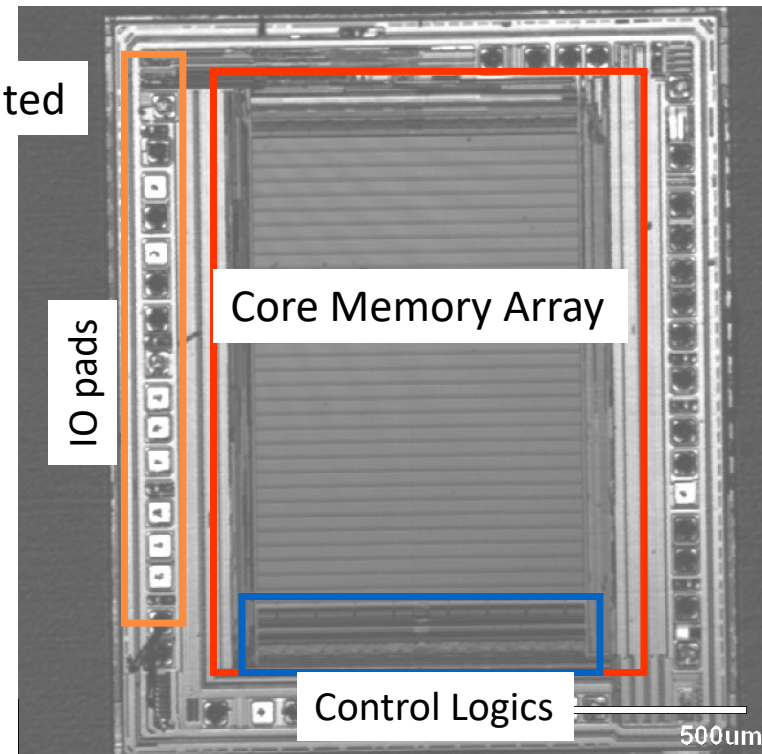
Commercial SRAM Chip
Cypress CY62128, 1 Mbit



Decapsulated



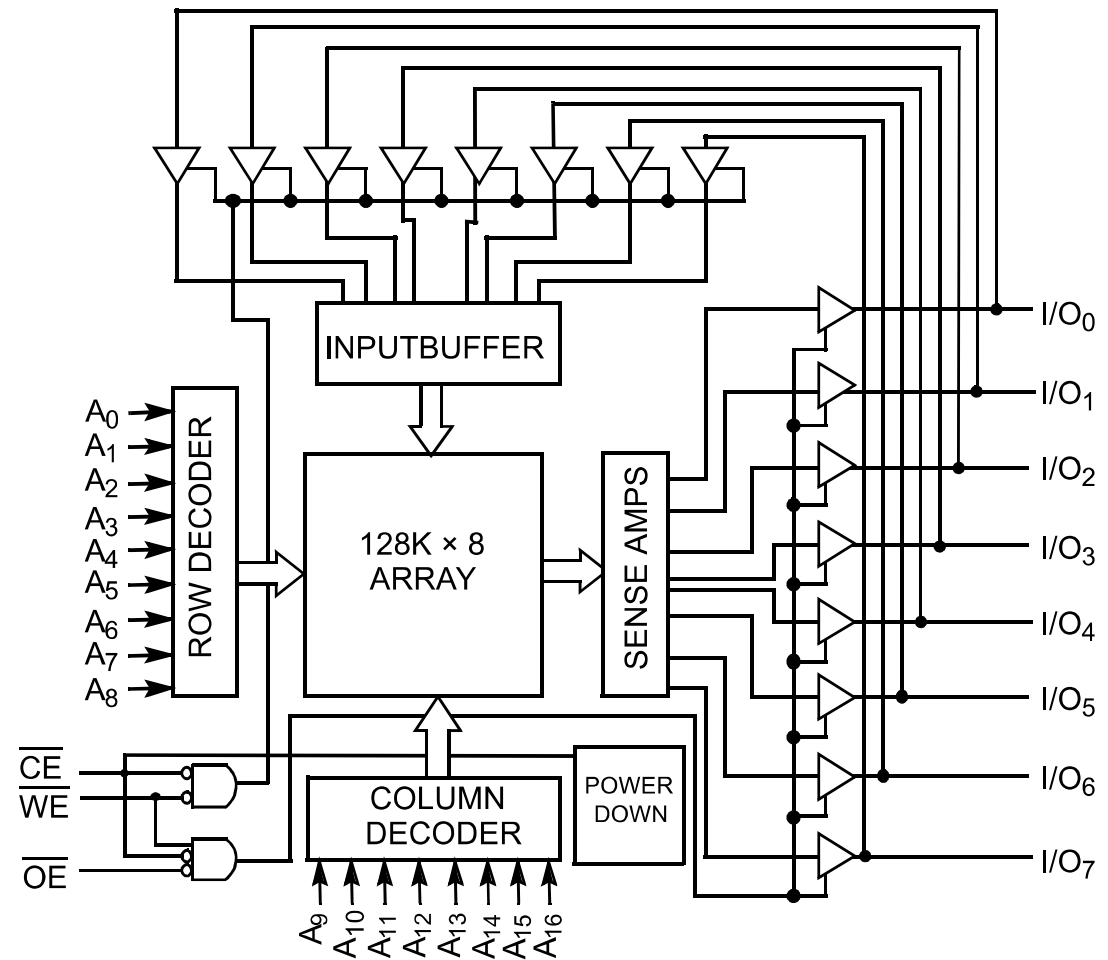
Decapsulated silicon die
Laser scanned image, 5x magnification



- **Memory** array takes up **most** of the **real estate**.
- Leading Edge Semiconductor Process is usually first tested with Memory Design.

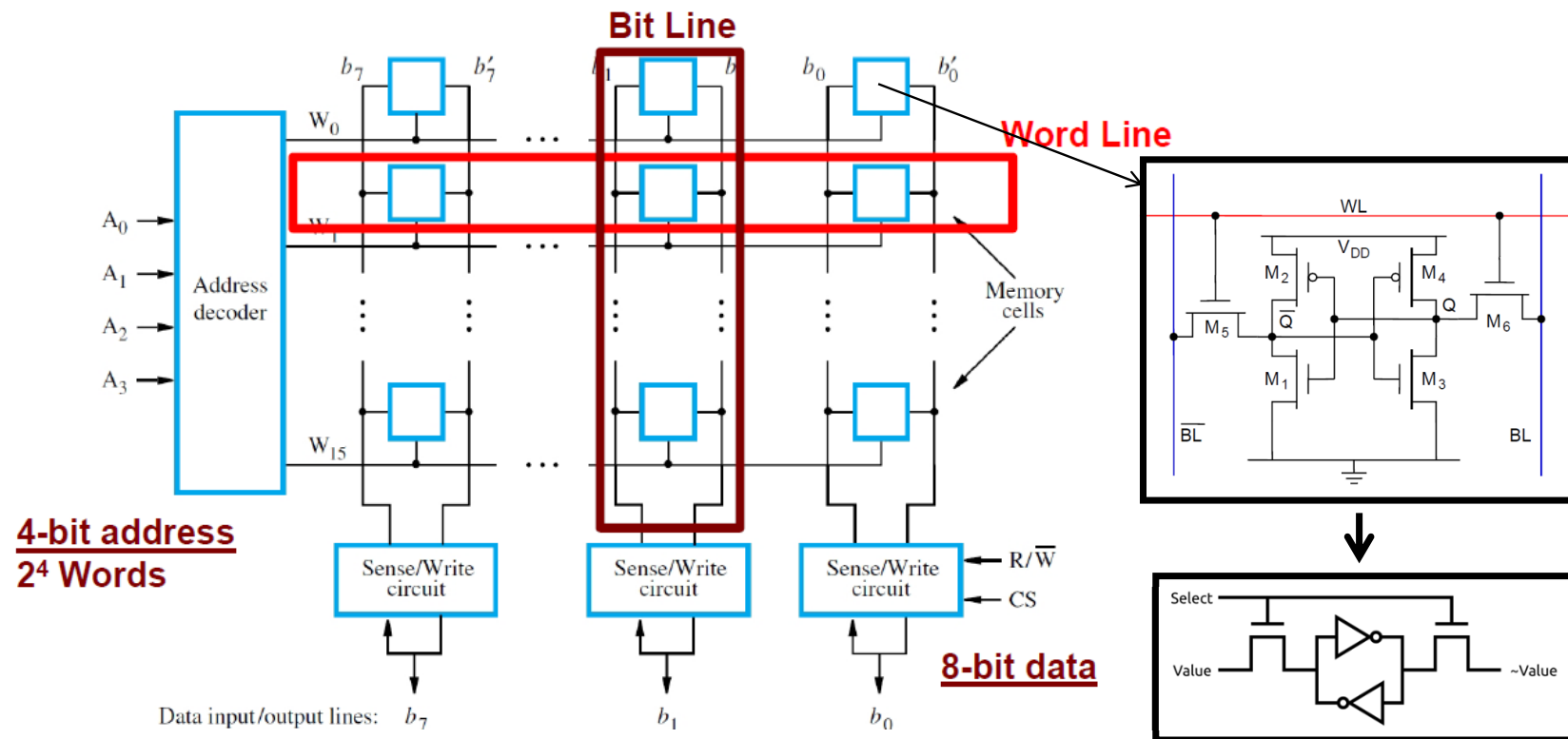
SRAM Internal Circuitry

- Memory array
- 1M SRAM cells for the chip shown.
- Control circuitry
- Decoders
- Sense amplifiers
- Input/Output multiplexers

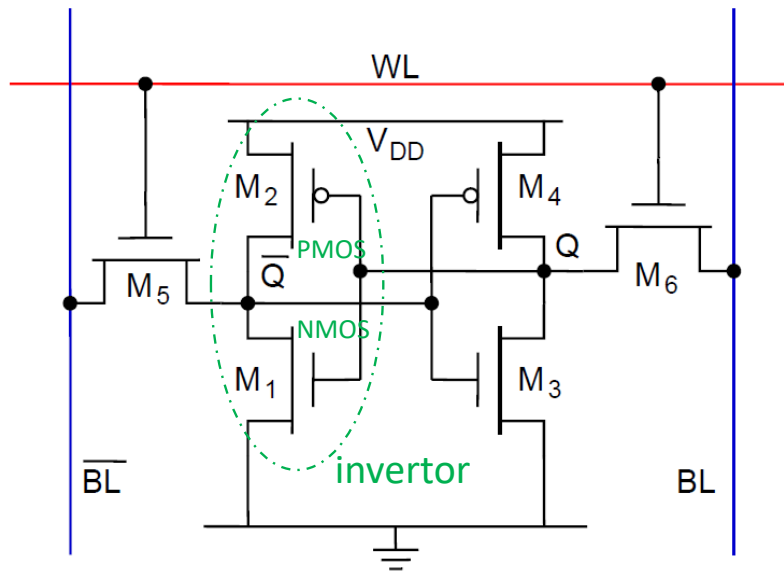


SRAM Cell

- Memory cells are organized in arrays (rows), and are accessed via the **word lines** and **bit lines**.
- Each individual SRAM Bit consist of 6 Transistors (typical design).

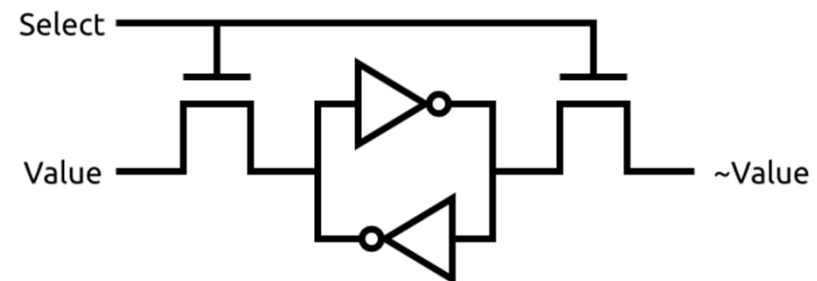


SRAM Cell

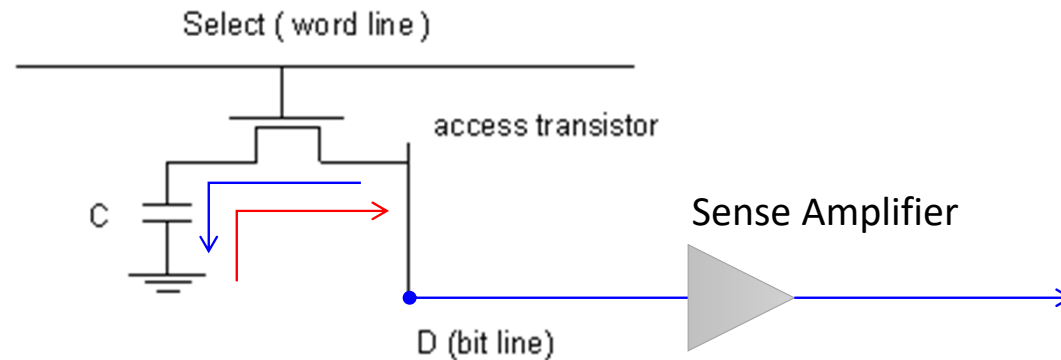


- The data bit is stored in M1, M2, M3 and M4 (which is equivalent to two inverters connected as shown on the right).
- M5 and M6 are pass (switch) transistors.

- There are six transistors
- M1, M2, M3, M4, M5 and M6
- Word line (WL) is derived from Address Decoder Output. It selects the word location for read and write
- The actual data are placed on the differential bit lines (BL and BLB). This is connected to the Data Bus.



Dynamic RAM (DRAM)



- Single Transistor Design
- DRAM uses a **capacitor** as its **storage** element.
- The Transistor is used to control charges flowing in and out of the capacitor during the Read and Write process.
- Write Process
 - To store a Logic '1': Enable the Transistor, transfer charge into capacitor.
 - To store a Logic '0': Enable the Transistor, discharge the capacitor.
- Read Process
 - Enable the Transistor. Measure the capacitor charge using a sense amplifier.



DRAM – Maintaining Data Integrity

- DRAM Read Process destroys information stored on capacitor
- The process of measuring charges on a capacitor also effectively discharged it i.e. data is destroyed.
- Hence, the original data has to be re-written back after every read.
- Periodic refresh is needed as the stored charge “leaks” with time.
- The basic DRAM is more or less obsolete in the market today. It is replaced by its synchronous version called Synchronous DRAM (SDRAM).
- Difference between SDRAM and DRAM is that the former makes use of a clock signal from the host to synchronize data transfer, enabling faster transfer rate. SDRAM also has a pipeline architecture that allow faster, overlapping operations.
- Other enhancements of SDRAM includes its double data rate versions DDR, DDR2, DDR3 SDRAM, which could reach transfer speed of more than 2G transfers per second.

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